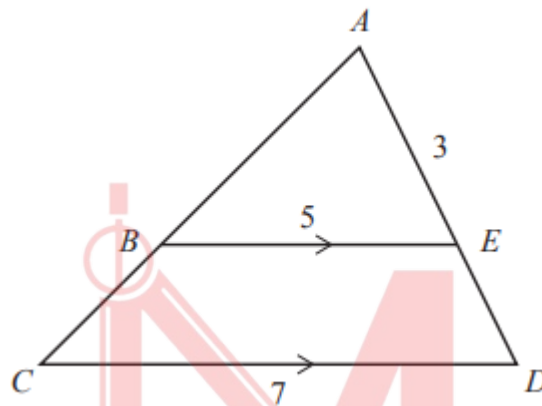


Similar Triangles

Worksheet 1 (Non-Calculator)

1. W14/qp12/q13



In the diagram, $BE = 5$ cm, $CD = 7$ cm and $AE = 3$ cm. BE is parallel to CD .

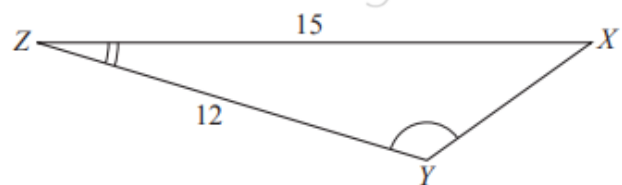
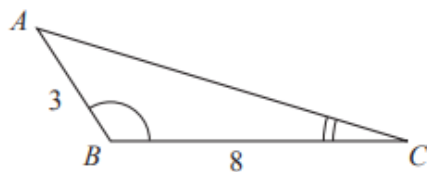
a. Express CD as a percentage of BE .

[1]

b. Find ED .

[2]

2. S15/qp11/q13



In the diagram, triangle ABC is similar to triangle XYZ . $AB = 3$ cm, $BC = 8$ cm, $YZ = 12$ cm and $ZX = 15$ cm.

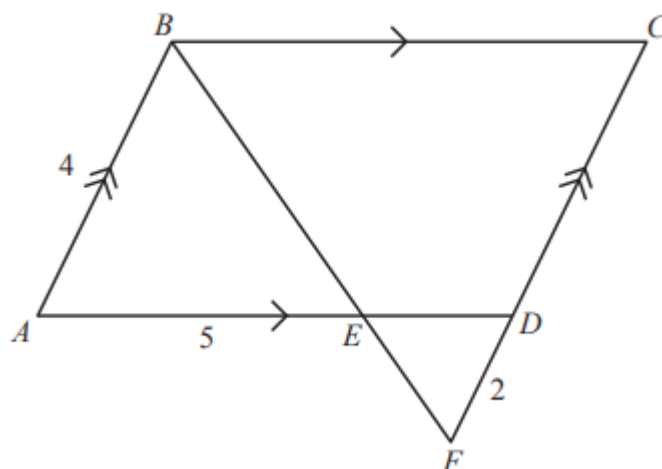
a. Calculate XY .

[1]

b. Given that the area of triangle ABC is 10 cm^2 , calculate the area of triangle XYZ .

[2]

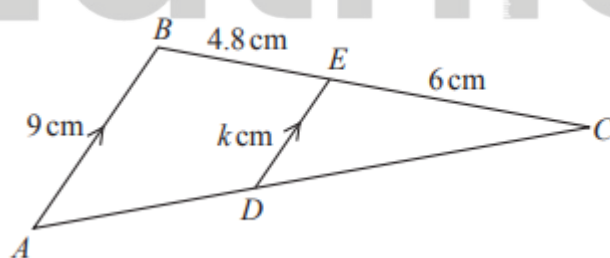
3. W15/qp11/q25



ABCD is a parallelogram. BEF and CDF are straight lines. $AB = 4\text{cm}$, $DF = 2\text{cm}$ and $AE = 5\text{cm}$.

- Show that triangles ABE and CFB are similar. Give reasons for each of your statements. [2]
- Calculate BC. [2]
- Triangle DFE is also similar to triangle ABE. Given that the area of triangle DFE is $x\text{ cm}^2$, find the area of ABCD in terms of x . [2]

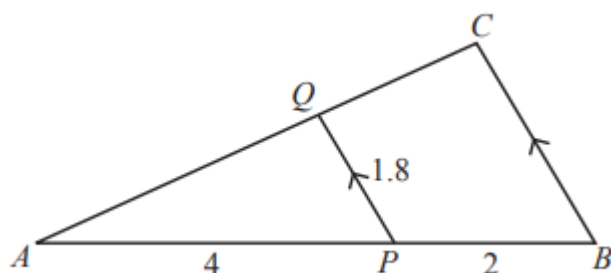
4. S16/qp23/q21(a) (0580)



NOT TO
SCALE

Triangles CBA and CED are similar. AB is parallel to DE. $AB = 9\text{ cm}$, $BE = 4.8\text{ cm}$, $EC = 6\text{ cm}$ and $ED = k\text{ cm}$. Work out the value of k . [2]

5. W16/qp11/q14



In the diagram, triangles APQ and ABC are similar. BC is parallel to PQ and AP = 4cm, PB = 2cm and PQ = 1.8cm.

a. Find BC. [2]

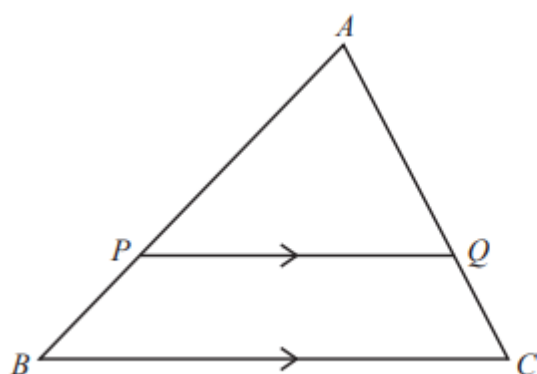
b. Find $\frac{(\text{area of triangle APQ})}{(\text{area of quadrilateral PBCQ})}$. [1]

6. S18/qp22/q11 (0580)



The diagram shows two mathematically similar triangles, T and U. Two corresponding side lengths are 3cm and 12 cm. The area of triangle T is 5cm^2 . Find the area of triangle U. [2]

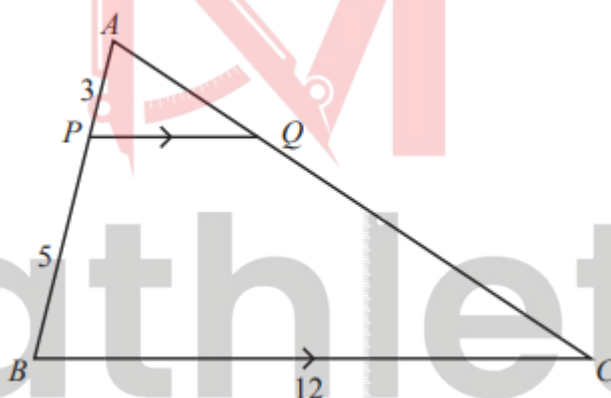
7. S18/qp23/q25 (0580)



NOT TO
SCALE

- a. In the diagram, PQ is parallel to BC. APB and AQC are straight lines. PQ = 8cm, BC = 10 cm and AB = 9cm. Calculate PB. [2]

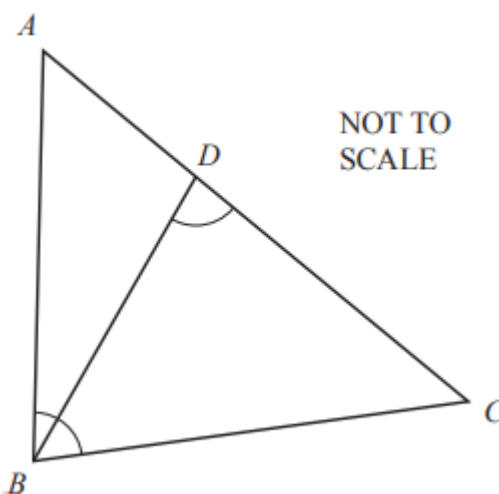
8. W18/qp11/q25



In the triangle ABC, P and Q are points on AB and AC such that PQ is parallel to BC. AP = 3cm, PB = 5 cm and BC = 12cm.

- a. Find PQ. [2]
b. The area of triangle ABC is $x \text{ cm}^2$. Find an expression, in terms of x , for the area of trapezium BCQP. [2]

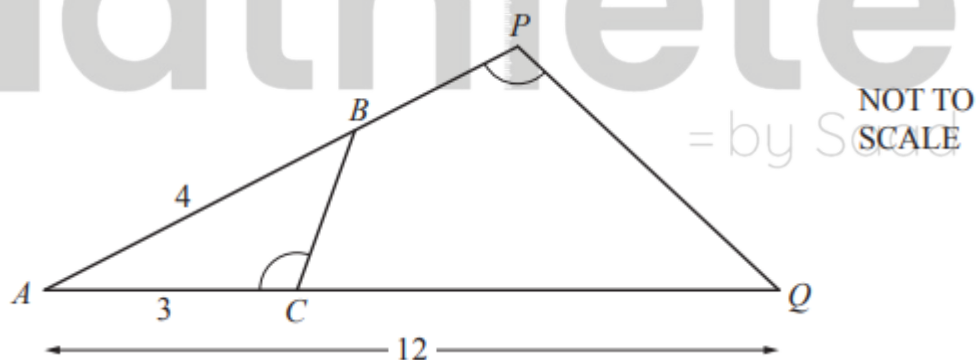
9. S19/qp11/q19



The diagram shows triangle ABC. D is a point on AC such that angle BDC = angle ABC.

- Explain why triangle BDC is similar to triangle ABC. Give a reason for each statement you make. [2]
- AC = 10 cm, BC = 8 cm and BD = 6 cm. Calculate AB. [2]

10. W20/qp11/q25



In the diagram, ABP and ACQ are straight lines.

$$\angle ACB = \angle BPQ.$$

- Show that triangle ABC is similar to triangle AQP. Give a reason for each statement you make. [2]
- AB = 4 cm, AC = 3 cm and AQ = 12 cm. Calculate AP. [2]
- The area of triangle ABC is $x \text{ cm}^2$. Find an expression, in terms of x , for the area of quadrilateral BPQC. [1]

Answer Key

1. W14/qp12/q13

13 (a)	140	1	
(b)	1.2	2	M1 for $3 \times \left(\frac{7}{5} - 1\right)$; or $3 \times \left(\frac{\text{their}(a)}{100} - 1\right)$; oe or a complete algebraic method.

2. S15/qp11/q13

13 (a)	4.5, or any equiv.	1	
(b)	22.5, or any equiv.	2	M1 for $10 \times \left(\frac{a}{b}\right)^2$, where a and b are corresponding sides, possibly cancelled down, with $a > b$.

3. W15/qp11/q25

25 (a)	Two corresponding pairs of angles equated, with reasons, from $\hat{BAE} = \hat{FCB}$ opp. angles of a parm. $\hat{ABE} = \hat{CFB}$ alternate angles $\hat{AEB} = \hat{CBF}$ alternate angles	2 *	B1 for any one pair, with correct reason
(b)	7.5 oe	2 *	M1 for $\frac{BC}{5} = \frac{6}{4}$ oe
(c)	12x	2 *	B1 for seeing 4x or 9x as $\triangle ABE$ or $\triangle BCF$ respectively

4. S16/qp23/q21(a) (0580)

21 (a)	5	2	M1 for $\frac{9}{k} = \frac{6+4.8}{6}$ oe
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5. W16/qp11/q14

14 (a)	2.7 oe	2 *	M1 for $\frac{BC}{6} = \frac{1.8}{4}$ oe
(b)	$\frac{4}{5}$ oe	1 *	

6. S18/qp22/q11 (0580)

11	80	2	M1 for $\left(\frac{12}{3}\right)^2$ or $\left(\frac{3}{12}\right)^2$ oe or $\frac{3^2}{5} = \frac{12^2}{A}$ oe
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7. S18/qp23/q25 (0580)

25(a)	1.8	2	M1 for $\frac{10}{8} = \frac{9}{AP}$ oe
25(b)	10.3 or 10.31 to 10.32	3	M2 for $13 \times \sqrt[3]{\frac{0.25}{0.5}}$ oe or M1 for $\sqrt[3]{\frac{0.5}{0.25}}$ oe or $\sqrt[3]{\frac{0.25}{0.5}}$ oe or $\frac{0.5}{0.25} = \left(\frac{13}{h}\right)^3$ oe

8. W18/qp11/q25

25(a)	4.5 oe	2	M1 for $\frac{PQ}{12} = \frac{3}{8}$ oe; or for $[PQ =] 12 \times \frac{3}{8}$
25(b)	$\frac{55}{64}x$ oe	2	B1 for $\left(\frac{3}{8}\right)^2$ or for $\left(\frac{8}{3}\right)^2$ seen or M1 for $[\text{area of } BCQP =] x - \text{area of } \triangle APQ$ provided area of $\triangle APQ$ is in terms of x .

9. S19/qp11/q19

19(a)	Angle C is common Angle $BDC = \text{angle } ABC$ (given) Hence similar	2	B1 for the two correct pairings or one pair with reason
19(b)	7.5 oe	2	M1 for $\frac{AB}{6} = \frac{10}{8}$ or $\frac{AB}{10} = \frac{6}{8}$ oe

10. W20/qp11/q25

25(a)	$\hat{ACB} = \hat{APQ}$ [given] $\hat{BAC} = \hat{QAP}$ same angle $\hat{ABC} = \hat{AQP}$ [third] angles in a triangle Hence similar	2	B1 for two correct pairs of angles identified = by Saad
25(b)	9 nfw	2	B1 for $\frac{AP}{3} = \frac{12}{4}$ oe
25(c)	8x	1	